Surfacing the European Idea in the Press of Interwar Romania

A Retrieval Funnel for Noisy, Unlabelled Historical Newspapers

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Interwar Romania and the idea of a united Europe

“What was the common people’s sentiment about a common Europe?”

The question that started our project. The way in: newspapers in Romanian, German and Hungarian, all scanned, none labelled.

OUR SCOPE IN THIS STUDY

Three German-language newspapers from Transylvania and the Banat, set in Fraktur. The Paneuropa programme called for protecting national minorities, and the historian on our team reads these papers natively.

KEY MOMENTS

1925

Coudenhove-Kalergi’s Paneuropa Union, founded in Vienna, calls for a “United States of Europe”.

Oct 1926

First Paneuropa congress in Vienna.

Sept 1929

Aristide Briand proposes a federal bond among the European peoples at the League of Nations.

The search is fast. The reading afterwards is not.



14,408

documents found by one search, “Paneuropa”, in the Arcanum archive, 1920–1939

The archive returns documents, not articles. Each document has many pages, and a page of the Siebenbürgisch-Deutsches Tageblatt holds many articles in several columns, so the historian reads every page to find the one that really discusses the topic.

Our tool changes only step 03. Search and later work stay as they are.

Two problems make the reading costly

01 — OCR noise hides the keyword

In our Fraktur pages, Paneuropa comes out as *baneuropa* or *Panzuropa*, Völkerbund as *bölferbund*.



- **32%** start with “paneuropa”: literal keyword
- **22%** other “paneurop…” forms: the stem too
- **20%** garbled: only the OCR-tolerant list
- **26%** too garbled for any keyword method

n = 120 “Paneuropa”-like tokens, 1,047 passages

02 — Keywords find words, not topics

Many hits name Europe only in passing; some articles argue for unity without the word. We tell three kinds apart:

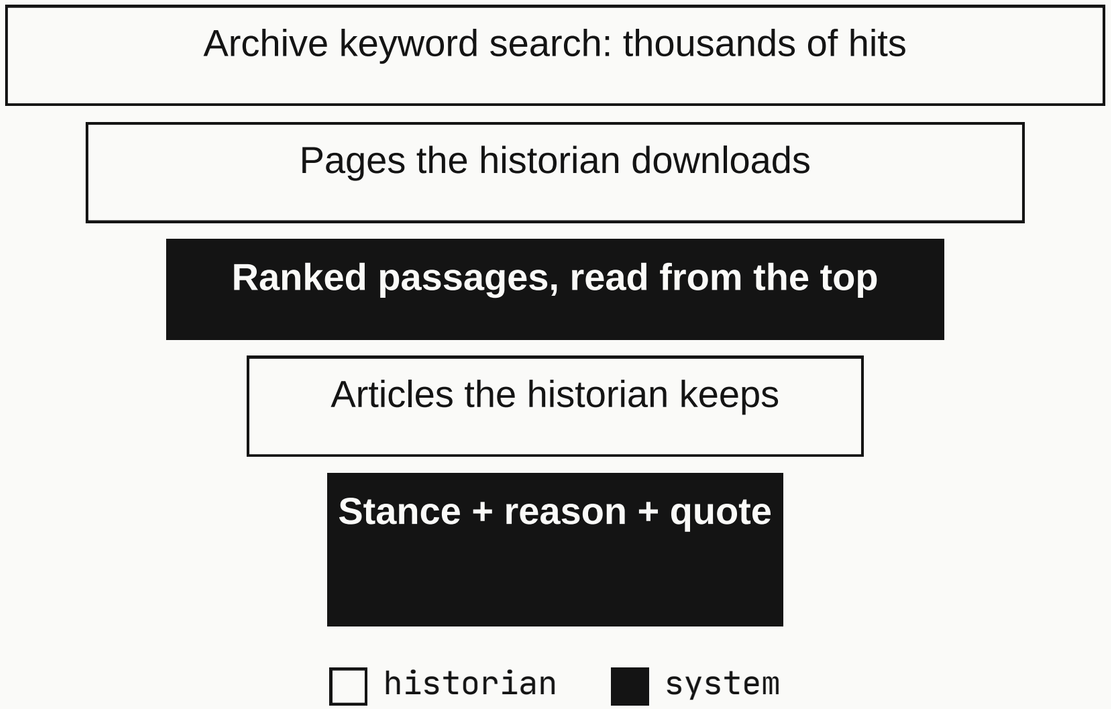
IDEA	The article discusses the idea of European unity.
MENTION	It names Europe or Paneuropa but is about something else.
implicit	An IDEA article that never uses the word “Europa”.

What historians asked for, and what we built

Gathered from researchers working on Paneuropa at a conference in Budapest, summer 2026

What historians need	Our design	In our study
Find the article, not every hit	Rank passages, count articles	tested
Find words the OCR misread	OCR-tolerant word list	tested
Find articles without the keyword	Search by meaning	tested
See why a passage came up	Matched term or nearest concept	shown, not tested
Reasons, not verdicts	Stance with a reason and a checkable quote	tested
Trace every finding to its page	Passage ids point to page and position	partly
Stance in the text’s own voice	Guideline rule written into the prompt	partly
Collapse reprints; show OCR quality	Not built yet	future
The rise of the debate, 1929–1932	Needs a longer run of pages	future

A funnel, not a classifier



WHY NOT CLASSIFY THE WHOLE ARCHIVE?

01 Relevant articles are rare, and we had no labels to train on.

02 A verdict on every page is hard for a historian to check.

03 The funnel ranks pages the historian already has: less reading, but still reading.

04 Labels are needed only to evaluate, not to train.

The human stays where the judgement lies.

From archive hit to checked stance

01 · HISTORIAN

Search and download

Keyword search in Arcanum; PDFs with OCR text layer

→

02 · SYSTEM

Cut into passages

Follow the columns, 40–250 words; OCR left as it is

→

03 · SYSTEM

Rank and explain

6 methods; each says why a passage came up

RQ1 · ranking

How much reading does it save?

→

04 · HISTORIAN

Read and keep

Read from the top; keep the articles on the idea

→

05 · SYSTEM

Stance with evidence

Open LLM: stance, reason, quote; quote checked

RQ2 · stance

Does it agree? Is the quote real?

The historian has the last word: agrees with or overrules every stance.

□ historian ■ system

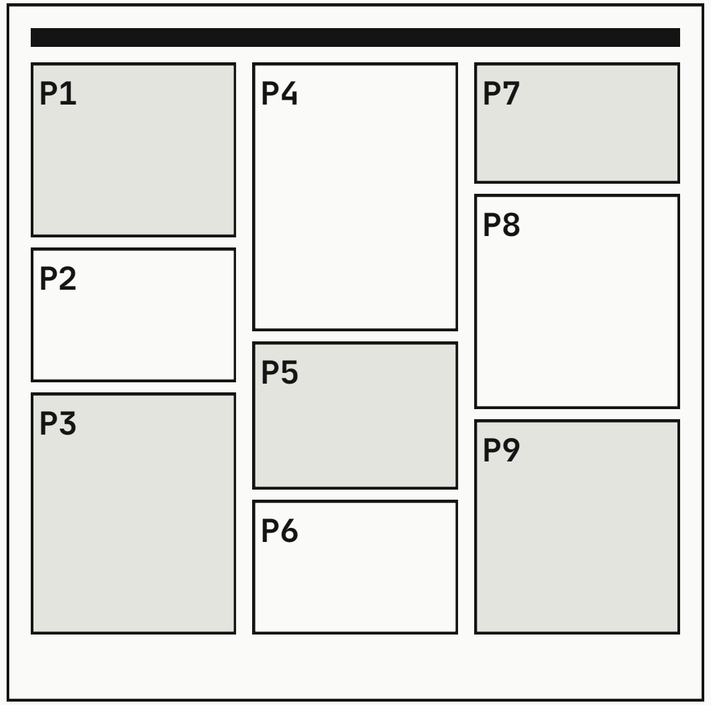
Step 02: cut pages into passages by column

The OCR does not mark where an article begins or ends, so the page is cut into passages first.

01	Read the archive’s OCR text layer, block by block.
02	Normalise only historical glyphs (long s, ligatures, quotation marks) and words split at a line end.
03	No OCR correction: every method faces the same noise as the archive search.
04	Follow each column top to bottom; merge short blocks until a passage has at least 40 words, never merging past 250.
05	Every passage points back to its page and position.

896 passages on 30 pages

52 words per passage (median)



One page, cut into passages along its columns (schematic)

Step 03: Ranking Passages

Method	Ranks passages by	Shown to the historian
R0	Reading order, top to bottom (no tool)	–
B0	Archive keyword: words starting with paneurop	the matched words
B0-literal†	Literal keyword: words starting with paneuropa	the matched words
B1	BM25 over our 13-term word list, exact spelling	the matched words
S1	BM25, same list, tolerant of OCR errors	OCR variant, e.g. bölferbund völkerbund
S2	Meaning: closeness to 3 concept descriptions	the closest description
S3	Rank fusion of S1 and S2	both explanations

BM25: rare words weigh more; a hit on völkerbund outweighs one on europa. Each step tests one effect: B1 vs B0 wider list · S1 vs B1 OCR tolerance · S2 meaning · S3 both.

S2 · THE 3 CONCEPT DESCRIPTIONS

1 Die Paneuropa-Bewegung und die Idee eines geeinten Europas: Zusammenschluss der europäischen Staaten zu einer politischen und wirtschaftlichen Union.

2 Europäische Zusammenarbeit und Verständigung zwischen den Völkern Europas, Frieden und eine gemeinsame Ordnung für Europa.

3 Vereinigte Staaten von Europa, europäische Föderation, europäische Zollunion, Europa gegenüber Amerika und Russland.

B1, S1 · WORD LIST

paneuropa · paneuropäisch · europa · europäisch · europäer · coudenhove · kalergi · briand · staatenbund · föderation · zollunion · völkerbund · verständigung

List and concept descriptions were fixed before any label existed. † added after first results, not pre-registered.

Showing the rank and a reason

THE ARTICLE · PAGE T1-06

A report of the 1926 Paneuropa congress in the Siebenbürgisch-Deutsches Tageblatt.



Der erſte paneuropäiſche Kongreß.

Wien, 5. Oktober.

Die geiſtige Tagung.

[…]

ſen des Geiſtes zu überzeugen, die Konflikte aufzulöſen und dadurch eine europäiſche Gemeinſchaft zu erzeugen. Napoleon hat auch zu Fouchées Aeußerungen getan, die darauf hindeuteten, er hätte ohne den Zuſammenbruch in Rußland den europäiſchen Völkerbund geſchaffen,

1 **paneuropäische**, OCR: paneuropäiiche. The keyword stem finds it: rank 24.

2 **europäische(n)**, OCR: europäifche, europäiidien. Only the OCR-tolerant list matches: rank 3.

SIEBENBÜRGISCH-DEUTSCHES TAGEBLATT · 12 OCT 1926 · ARCANUM

OCR tolerance pays off: rank 3 instead of 291.

Method	Rank	Why it came up
S1 OCR-tolerant list	3	europäifche europa
S3 fusion (S1 + S2)	3	europäifche europa
S2 meaning	12	close to concept 2, European cooperation
B1 exact list	20	europäer
B0 archive keyword	24	paneuropäiiche
B0-literal	291	no match, reached by reading on

Rank of the article's best passage among all 608 passages of the keyword pages (1 = read first)

Dataset for RQ1

Page set	Selected by	Pages	Passages	IDEA	MENTION
Keyword pages	Arcanum search for “paneuropa”	20	608	15	6
Random control	random draw, no search	10	288	1	0

Siebenbürgisch-Deutsches Tageblatt 10/1926 · Temesvarer Zeitung 01/1926 · Banater Deutsche Zeitung 08/1929 · all Fraktur

GOLD STANDARD

01 Marked by a historian on the team, blind to all system output

02 The topic decides, not the word: implicit articles count

03 The article is the unit: one grade, one stance

04 One annotator: agreement not measured

MAIN MEASURE: READING EFFORT

E80 = passages read from the top until 80% of the IDEA articles are found

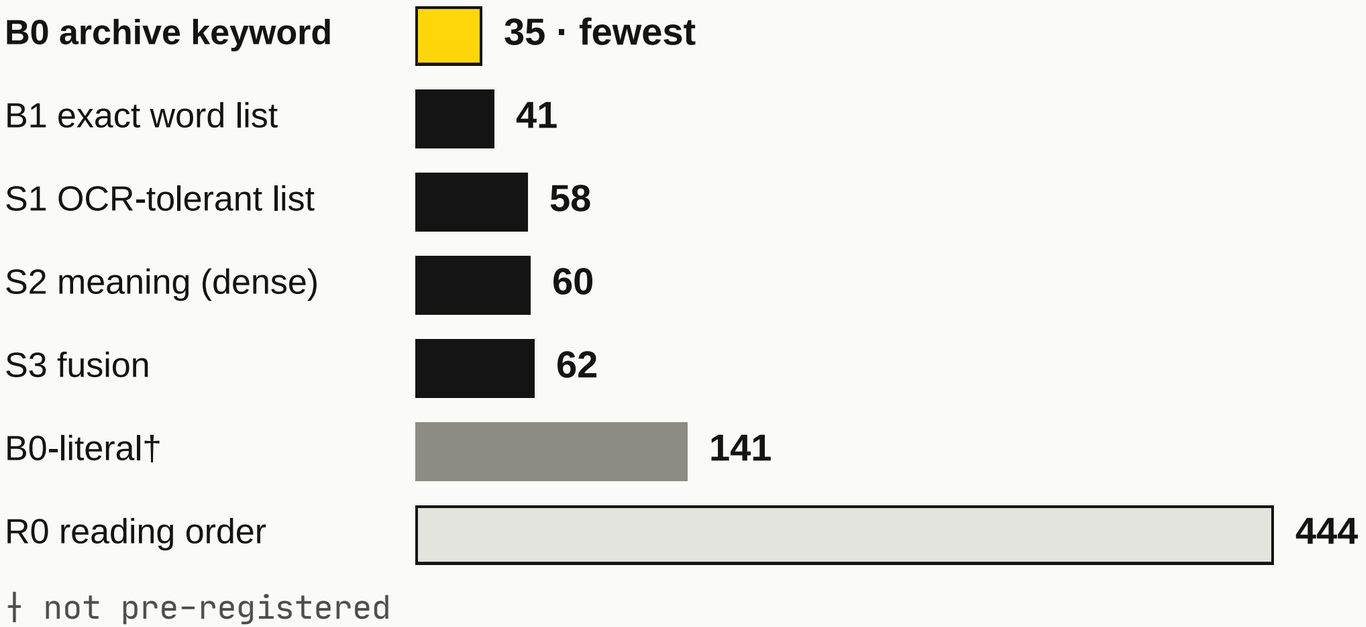
An article counts as found once any of its passages is read. After Cooper’s expected search length. Also reported: recall at 10 and 20, implicit articles, nDCG@10.

RESULTS RQ 1

How much reading does ranking save?

Ranking cuts reading from 444 to 35 passages

E80: passages read until 80% of the 15 IDEA articles are found · 608 passages · lower = less reading



† not pre-registered

×7

less reading or better, with any ranking method

Home advantage

for the keyword: pages from its own search, good OCR. Untested on open archives.

0.93

nDCG@10 for search by meaning: the cleanest top ten of the pre-registered methods

RESULTS RQ 2

Can a model suggest a stance, with a quote to check?

Step 05: a stance with a reason and a quote

SETUP

01	Qwen3-30B-A3B (instruct 2507), open weights
02	Zero-shot, temperature 0, one run
03	Run at GWDG (SAIA); prompts not stored
04	Reads the whole article, never the annotator's label

AUTOMATIC QUOTE CHECK

exact	occurs as it stands
close	fuzzy match ≥ 90 of 100
not found	otherwise

THE KEY RULE IN THE PROMPT

"The voice that counts is the text's, not that of the people it quotes: a neutral report of a pro-Europe speech is still NEUTRAL."

WHAT THE MODEL MUST RETURN

```
{
  "stance": "PRO" | "CONTRA" | "NEUTRAL",
  "reason": "<one sentence>",
  "quote": "<from the article, ≤ 40 words>"
}
```

The model agrees on 7 of 15 stances

Historian ↓ / model →	PRO	CONTRA	NEUTRAL
PRO (5)	5 ✓	0	0
CONTRA (1)	0	0 ✓	1
NEUTRAL (9)	7 ✗	0	2 ✓
UNSURE (1)	0	1	0

✓ agreement · Cohen’s κ = 0.13 · UNSURE reported apart

It finds every PRO article, but reads 7 of 9 NEUTRAL ones as PRO.

In review, 3 of the 8 disagreements turned out defensible: part of the gap lies in the task definition.

01 **Reported speech taken as the paper’s voice**

3 congress reports summarise pro-European speeches; the model adopts the speakers’ view.

02 **Reconciliation read as the European idea**

4 articles on French–German rapprochement; for the historian, that is not yet the European idea.

03 **Quotes: 5 exact · 7 close · 4 not verbatim**

The 4 join pieces with “…” or fix OCR silently (*Interesse* for *iyereffe*). None looks invented.

What it means, and what we did not build

FOR HISTORIANS

01 Holding keyword pages? Read the keyword passages first.

02 Beyond them, word lists and meaning surface what a search never returned (one article, on random pages).

03 The implicit debate still needs reading.

04 The model’s stance is a pointer to check, not a verdict.

FROM PLAN TO SLICE

01 Planned: four stages, 1918–1939, Romanian, German and Hungarian press.

02 Built: ranking and stance with evidence, on 30 pages.

03 Why: no mass downloads, subscription ended; no gold standard existed.

LIMITATIONS

01 Pages come from Arcanum's keyword search; we can't see what it missed.

02 One annotator, no agreement measured.

03 Word list and concepts written by us.

04 Three German titles; year and paper can't be separated.

05 Passages, not minutes: no user study yet.

Ranking saves reading. A model may suggest a stance, but only with a quote the historian can check.

OPEN QUESTIONS

01 Do fewer passages mean less time and more trust?

02 Does it hold for the Romanian- and Hungarian-language press?

03 Would a second historian mark the same articles and stances?

04 How can we find what the archive search never returned?

WHAT COMES NEXT

We are turning the funnel into a tool historians can use, and testing it with them. We are looking for historians and collections to try it on.

`github.com/tanruoshan/popline`

The newspaper pages are subscription material and not included.

Thank you!